Computer Network

3rd September 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

a) Compare the OSI model structure, with the TCP/IP model in computer network. You should clearly state the function of each layer and how the layers in the two models correspond. **10**

© NCC Education Limited 2024

Marks

Mark Scheme

TCP/IP is a concise version of OSI model where OSI model consists of 7 layers such as Application Layer, Presentation Layer, Session Layer, Transport Layer, Network Layer, Data Link Layer and Physical Layer whereas TCP/IP model consists of 4 layers such as Application Layer, Transport Layer, Network Layer and Network Interface Layer.

1 mark for stating OSI is 7 layers and TCP/IP is 4 layers. 1 mark for listing down all the 7 layers of OSI without error and 1 mark for listing down all the 4 layers of TCP/IP without error.

Each layer of OSI and TCP/IP has a certain function to it:

<table>
<tr><th>OSI Model</th><th>TCP/IP Model</th></tr>
<tr><td>Application Layer provides services to end-users, such as file transfer, web browsing and email, etc.</td><td rowspan="3">Application Layer is the combination of the Application, Presentation and Session Layer of the OSI model. Application Layer indicates the applications and programs that utilises the TCP/IP model for communicating with the user through applications and perform various tasks including data representation for the applications executed by the user and forwards it to the transport layer.</td></tr>
<tr><td>Presentation Layer is for converting data from one format to another so that it can be understood by both endpoints on a network.</td></tr>
<tr><td>Session Layer responsible for creating, maintaining and managing the communication between two end points on a network.</td></tr>
<tr><td>Transport Layer responsible for providing reliable end-to-end delivery of data packet across a network.</td><td>Transport Layer is similar to the transport layer of the OSI model. It works for reliability and error-free message delivery.</td></tr>
<tr><td>Network Layer responsible for routing data packet from one node to another on a network.</td><td>Network Layer is similar to the network layer of the OSI model. It helps in the movement of data packet from the destination to the source.</td></tr>
<tr><td>Data Link Layer provides reliable data transmission over a physical communication link.</td><td rowspan="2">Network Interface Layer comprises of Physical and Data Link Layer of the OSI model. It has to work of transmitting the data among two devices in the network.</td></tr>
<tr><td>Physical Layer is to establish, maintain and terminate communication between two endpoints on different networks.</td></tr>
</table>

1 mark each for comparing each layer of OSI layers to TCP/IP layers. Maximum 7 marks for the layer to layer comparison.

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 2

a) List **FOUR (4)** network topologies used in Ethernet and fast Ethernet and explain each of the topologies. Name **ONE (1)** topology which is becoming obsolete **and** explain the reason behind it. **10**

Mark Scheme

- ***Bus topology is a network configuration where all devices are connected to a single central cable or bus. (1 mark is for correct identification and 1 mark for correct description).***
- ***Star topology is a network configuration where all devices are connected to a central hub or switch. (1 mark is for correct identification and 1 mark for correct description).***
- ***Hybrid topology is a network configuration that combines two or more different topologies. (1 mark is for correct identification and 1 mark for correct description).***
- ***Tree topology is a hierarchical network configuration resembling a tree structure, with nodes connected in parent-child relationships. (1 mark is for correct identification and 1 mark for correct description).***
- ***Bus topology becoming obsolete due to limited scalability, susceptibility to collisions and difficulty in fault detection. (1 mark is for correct identification of the obsolete topologies and 1 mark is for the reasons).***

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 3

a) Draw a network logical *diagram* to represent the following scenario. **10**

You are tasked with designing a network diagram for a small LAN (Local Area Network) in an office environment. The office consists of three departments: Sales, Marketing and Administration, each with its own set of requirements and devices.

Requirements:
1. Sales Department:
   - 10 desktop computers for sales representatives
   - 1 printer shared among sales representatives
   - Access to a shared sales database hosted on a server

2. Marketing Department:
   - 8 desktop computers for marketing executives
   - 1 printer shared among marketing executives
   - Access to design software installed on a dedicated workstation

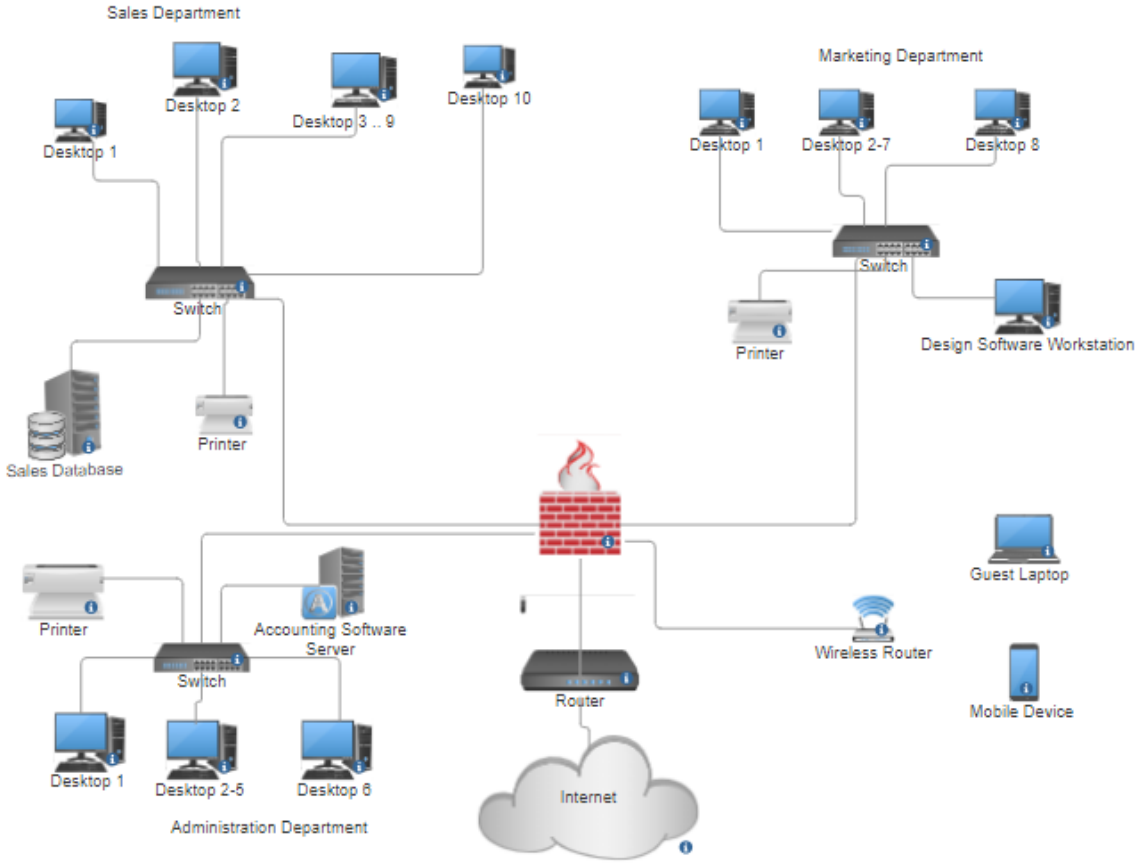
3. Administration Department:
   - 6 desktop computers for administrative staff
   - 1 printer shared among administrative staff
   - Access to accounting software installed on a dedicated server

Additional Information:
- All departments require internet access connection.
- Wireless access is needed in common areas for guests and mobile devices.
- Security measures must be implemented to safeguard data and network resources.
- Scalability and flexibility for future expansion should be considered in the design.

© NCC Education Limited 2024

Marks

Mark Scheme



- ***Award 1 mark for the Sales Department which contain 10 desktops, 1 shared printer, 1 switch and 1 sales database. All devices must connect to switch (1 mark)***
- ***Award 1 mark for the Marketing Department which contain 8 desktops, 1 shared printer, 1 switch and design software workstation. All devices must connect to switch (1 mark)***
- ***Award 1 mark for the Administration Department which contain 6 desktops, 1 shared printer, 1 switch and accounting software server. All devices must connect to switch (1 mark)***
- ***1 mark for wireless connectivity for the common area with Wireless Access Point (WAP)***
- ***1 mark if WAP connected to FW rather than a switch (better security)***
- ***1 mark for security measurement i.e. including a firewall***
- ***1 mark for correct position of FW before the router connecting to the internet to safeguard the LAN.***
- ***Award marks if servers consolidated into one place in LAN***

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 4

a) Justify the usage of the **FOUR (4)** network components used in your answer in the Question 3a. **4**

Mark Scheme

- ***A switch connects multiple devices on a computer network.***
- ***A router is used to connect a LAN with an internet connection.***
- ***A wireless access point provides wireless access to the internet or to the LAN.***
- ***Servers such as the accounting software server that provide the accounting application for the administration department.***
- ***Firewall for security***

Award 1 mark each for any bullet point above or any valid alternative answers.

b) You have been tasked with troubleshooting a network issue reported by users in the Sales department. Users are experiencing intermittent connectivity problems and slow network performance when accessing the shared sales database hosted on a server. **6**

Outline **THREE (3**) types of information you will gather about the problem to start your troubleshooting. You should specify the source of information and/or any tools you use to gather this information.

Mark Scheme

Types of information to be gathered can be:

- ***Problem timing and symptoms which can be gathered from system users through interview. (1 mark for information and 1 for the source).***
- ***Network performance such as response time between user request and server response which can be gathered from network monitoring tools such as real time data from Wireshark and database servers. (1 mark for information and 1 for the source).***
- ***System logs such as error messages and user log ins from firewall, switches and router log files. (1 mark for information and 1 for the source).***
- ***Security incidents from firewall, IDS, router and switches. (1 mark for information and 1 for the source).***

Award 2 marks each for any of the 3 bullet points or any valid alternative answers.

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 5

a) Explain **THREE (3)** factors you should consider when deciding on the choice of network media. **6**

Mark Scheme

Possible answers:

- ***Cost and Installation: evaluate the cost of the media itself and the installation process. Some media options may be more expansive upfront or require specialised equipment.***
- ***Bandwidth rating: when selecting network media, considering the bandwidth rating is crucial. The bandwidth rating determines the data transfer capacity of a medium and how quickly data can be transmitted.***
- ***Maximum segment length: refers to the farthest distance a signal can travel along a specific type of cable without requiring a repeater or signal booster.***
- ***Environment: Consider the environment where the media will be installed. Is it indoors or outdoors? Will it be exposed to extreme temperatures, moisture or other environmental factors? Different media have varying levels of durability and resistance to environmental conditions.***

Any 3 from the above list to be accepted,1 mark for correct identification and 1 mark for the explanation.

b) Compare and contrast CAT5e and CAT6a cabling in a LAN with respect to the following factors: **4**

1. Bandwidth
2. Speed
3. Structure
4. Application Area

© NCC Education Limited 2024

Marks

Mark Scheme

1) ***Bandwidth: CAT 5e 100 MHZ and CAT 6a 250 MHZ (1 mark, if the student only stated CAT 6a has a greater bandwidth compare with CAT5e, no mark awarded)***
2) ***Speed: Both CAT5e and CAT 6a supporting speed up to 1 Gbps (1 mark)***
3) ***Structure: Both CAT5e and CAT6a cables use twisted pair design. CAT6 has more stringent specifications for reducing crosstalk and system noise compared to CAT5e. (1 mark)***
4) ***Application: CAT5e: Suitable for basic LANs, home networks and office environment. CAT6a: ideal for future-proofing networks, high speed data transfer and demanding applications like video streaming and large file transfers (1 mark)***

Total 10 Marks

Question 6

a) Imagine a scenario where a small office network consists of a single router connecting multiple computers to the internet. One day, the router fails, causing a complete loss of internet connectivity for all users in the office.

i) Explain which principle of network design can provide the solution for this issue. Give a justification for your answer. **2**

Mark Scheme

The principle that can resolve the issue is redundancy (1 mark). By introducing redundancy, the network can continue functioning even if a critical component fails (1 mark).

ii) Outline **ONE (1)** specific device that could be implemented to provide the solution in the network and describe how it would help mitigate the impact of the router failure. **2**

Mark Scheme

One specific method to achieve this is by implementing a secondary or backup router (1 mark). Set up a secondary router alongside the primary router, both routers should be connected to the same LAN and have access to the internet. If one router fails, the other handles all traffic thus preventing downtime (1 mark).

© NCC Education Limited 2024

Marks

b) Explain the following computer network threats and state **ONE (1)** example for each of the security features or controls which can detect/mitigate them. **6**

- DOS
- Packet sniffing
- Phishing

Mark Scheme

- ***DOS stands for Denial-of-Service. A DOS attack aims to overwhelm a network, server or service by flooding it with excessive traffic or requests, resulting in legitimate users being unable to access the resources. IDS (Intrusion Detection System) or IPS (Intrusion Prevention System) can filter and block suspicious traffic patterns. (1 mark for the explanation and 1 mark for the security control)***

- ***In a packet sniffing attack, hackers intercept and analyse unencrypted network traffic to extract sensitive information such as login credential and personal data. Encrypting network traffic using protocol like HTTPS or SSH and use Virtual Private Networks (VPNs) to create an encrypted tunnel, preventing packet sniffers from accessing sensitive data. (1 mark for the explanation and 1 mark for the security control)***

- ***Phishing involves tricking users into revealing sensitive information such as login credentials or financial details by posing as a trustworthy entity via email, messages or fake websites. Educate users about phishing risks, implementing email filters to detect suspicious messages and multi-factor authentication (MFA) adds an extra layer of security by requiring additional verification beyond passwords which can mitigate phishing. (1 mark for the explanation and 1 mark for the security control)***

Total 10 marks

© NCC Education Limited 2024

Question 7 **Marks 10**

Below is a Risk Assessment Matrix which can be used to identify the risk level from likelihood and impact levels of a threat.

Use the matrix to identify the overall risk score for the threats shown in the table below and devise **ONE (1)** control (or resolution strategy) each identified risk to mitigate its likelihood and reduce its impact on the network infrastructure.

Risk Matrix		Severity				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	**Almost Certain**	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikely	Low	Low	Medium	Medium	High
	Rare	Low	Low	Low	Low	Medium

Risk	***Likelihood***	***Impact***	***Risk Score***	***Control***
Unauthorised access	Possible	Minor		
Insider threat	Likely	Moderate		
Hardware failure	Possible	Moderate		
Power outage	Rare	Minor		
Human error	Likely	Major		

© NCC Education Limited 2024

Marks

Mark Scheme

Risk	***Likelihood***	***Impact***	***Risk Score***	***Control***
Unauthorised access	Possible	Minor	***Medium (1)***	***2-factor authentication technology (1)***
Insider threat	Likely	Moderate	***High (1)***	***Network monitoring (1)***
Hardware failure	Possible	Moderate	***High (1)***	***Hardware backup (1)***
Power outage	Rare	Minor	***Low (1)***	***Backup generator (1)***
Human error	Likely	Major	***Very high (1)***	***Staff training (1)***

Award 1 mark for each correct answer for the risk score and 1 mark for each correct answer for the control.

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 8

a) Name and explain **TWO (2)** different types of firewall and compare and contrast them in respect of their principles of operation, advantages and problems. **10**

Mark Scheme

Packet Filtering Firewall (1 mark)

- ***is a network security device that filters incoming and outgoing network packets based on a predefined set of rules (1 mark).***
- ***Validation element: operates at the network layer and evaluates packet attributes like source IP, destination IP, ports, and protocols (1 mark)***
- ***Advantage: Fast processing due to simple rule-based filtering (1 mark).***
- ***Problem: Susceptible to spoofing, can be tricked by forged IP addresses (1 mark).***

Application Gateway (1 mark)

- ***Also known as proxy server, is a server program that understands the type of information being transmitted (1 mark).***
- ***Validation Element: Operates at the application layer (1 mark).***
- ***Advantage: Enhanced security blocks malicious contents effectively (1 mark).***
- ***Problem: Performance impact, slower due to content analysis (1 mark).***

Maximum 10 marks

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 9

a) Answer the following questions about a VPN.

i) Explain what VPN stands for. Discuss **TWO (2)** primary functions of VPN and provide **ONE (1**) example of a scenario where using a VPN is particularly beneficial. **5**

Mark scheme

VPN stands for Virtual Private Network (1 mark), which is an encrypted connection over the internet from a device to a network (1 mark).

VPN enhances security through encryption (1) and enables access to geographically restricted content (1 mark).

Example:

- ***Use a VPN to secure data when using public Wi-Fi (1 mark)***
- ***Access region specific content while traveling (1 mark)***

1 mark for any bullet point listed above. Max 1 mark.

ii) Which **TWO (2)** CIA properties does VPN provide in communication? **2**

Mark scheme

Confidentiality (1 mark), integrity (1 mark)

iii) Name **ONE (1)** VPN protocol. **1**

Mark Scheme

- ***IPSec***
- ***PPTP***

Max 1 mark

b) State **TWO (2)** basic functions of a VOIP. **2**

Mark Scheme

- **Signalling (1 mark)**
- **Database service (1 mark)**
- **Call connect and disconnect (1 mark)**
- **Codec operation (1 mark)**

1 mark for any bullet point listed above. Max 2 marks.

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 10

a) Answer the following questions about IPv4 addressing.

i) Explain IP addressing classes and their usage in different size networks. **8**

Mark Scheme

- ***CLASS A starts with 0.0.0.0 and goes up to 127.255.255.255 (1 mark). Suitable for large networks (major corporations, ISPs) (1 mark).***
- ***CLASS B starts with 128.0.0.0 and goes up to 191.255.255.255 (1 mark). Class B IP addresses are used for medium-size networks (1 mark).***
- ***CLASS C starts with 192.0.0.0 and goes up to 223.255.255.255 (1 mark). Commonly used for smaller networks (home networks, small businesses) (1 mark).***
- ***CLASS D starts with 224.0.0.0 and goes up to 239.255.255.255 (1 mark). Reserved for multicast groups (using for streaming data to multiple recipients) (1 mark).***

ii) Explain subnetting and its advantage in networking. **2**

Mark scheme

Subnetting is a method of dividing a single physical network into logical sub-networks (subnets) (1 mark). Subnetting improves network security by isolating different parts of the network (1 mark).

Total 10 marks

END OF PAPER

© NCC Education Limited 2024

Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1	Yes
2	1,2	Yes
3	2	Yes
4	4	Yes
5	4	Yes
6	3,4	Yes
7	3	Yes
8	3	Yes
9	5,6	yes
10	1	yes

Grade Descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand network and communication protocols	Has little to no awareness of principles and concepts underlying of network and approaches and is unable to identify associated protocols.	Has vague awareness of principles and concepts underlying of network and approaches and is only superficially able to identify associated protocols.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated protocols.	Has very good awareness of principles and concepts underlying frameworks and approaches and is able in detail to identify associated protocols.	Has an excellent awareness of principles and concepts underlying network and approaches and is comprehensively able to identify associated protocols.
Understand the principles of common network topologies and architectures	Has little to no awareness of principles and concepts underlying network topologies and approaches	Has vague awareness of principles and concepts underlying network topologies and approaches and is only	Has satisfactory awareness of principles and concepts underlying network topologies and approaches	Has very good awareness of principles and concepts underlying network topologies and approaches	Has an excellent awareness of principles and concepts underlying network topologies and approaches and is comprehensiv

© NCC Education Limited 2024

	and is unable to identify associated architecture.	superficially able to identify associated architecture.	and demonstrates sufficient ability to identify associated architecture.	and is able in detail to identify associated architecture.	ely able to identify associated architecture.
Understand how to design and develop a network system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Understand the application of network security measures	Demonstrates little to no ability to adequately use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use security principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use security principles to effectively develop an artefact to solve an identified issue.
Be able to select and configure the hardware components of a computer network to meet the requirements of a precise specification	Demonstrates little to no ability to adequately develop a network using hardware components.	Demonstrates an extremely limited ability to adequately develop a network using hardware components.	Demonstrates a satisfactory ability to develop a network using hardware components.	Demonstrates a very good ability to develop a network using hardware components.	Demonstrates an excellent ability to develop a network using hardware components.
Be able to define and configure internet	Demonstrates little to no ability to adequately	Demonstrates an extremely limited ability	Demonstrates a satisfactory ability to	Demonstrates a very good ability to define and	Demonstrates an excellent ability to define and

© NCC Education Limited 2024

telephony and communication systems	define and configure internet telephony and communication systems.	to adequately define and configure internet telephony and communication systems.	define and configure internet telephony and communication systems.	configure internet telephony and communication systems.	configure internet telephony and communication systems.

© NCC Education Limited 2024